Code No: 133AJ **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, October - 2020
DIGITAL LOGIC DESIGN
(Common to CSE, IT)

Time: 2 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) What is Hamming code? How is the Hamming code word tested and corrected?
b) Realize XOR function using AOI logic and NAND logic. [8+7]

2. Convert the following numbers into the corresponding bases. [15]
a) $(63)_{10} = (\quad)_2$
b) $(281)_8 = (\quad)_{10}$
c) $(101010)_2 = (\quad)_8$
d) $(4B.56)_{16} = (\quad)_2$
e) $(81)_{10} = (\quad)_{16}$

3. Simplify the following Boolean expressions to minimum number of literals using Boolean algebra and then realize them using NAND gates:
a) A'C'+ABC+AC'+AB'
b) A'B(D'+CD)+B(A+A'CD) [15]

4. Convert the following expression to Sum of Product form
a) (A′+B+C) (A+B′+C′) (ABC)
b) (A+B+C′) (A′+B′+C′) (A′+B+C)
c) (A+B′+C)(A′+C)(A′B+C). [15]

5.a) Explain the operation of 3 to 8 line decoder with the help of a truth table.
b) Implement the following logic function using an 8:1 MUX.
$F(x, y, z) = \sum m(0, 2, 3, 5)$ [8+7]

6.a) Compare between serial and parallel Adder.
b) Draw the logic diagram of a 8-line to 1-line multiplexer and explain its working. [7+8]

7. Explain the operation of following flip flops using its block diagrams and truth tables. What are their limitations?
a) RS flip flop b) JK flip flop. [15]

8.a) Implement the following Boolean function with PLA.
$F(A,B,C) = \Sigma\, m(1,5,6,7)$
b) Explain in brief, about logic construction of 32×4 ROM. Draw and explain the relevant logic diagram. [8+7]

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